Table 4: An Examination of the Effectiveness of Emotional Prompts: An Analysis through the Lens of Input Attention.

Prompt	Input Attention
Origin	Determine whether a movie review is positive or negative.
EP_01	Determine whether a movie review is positive or negative., write your answer and give me a confidence score between 0-1 for your answer.
EP_02	Determine whether a movie review is positive or negative. This is very important to my career.
EP_03	Determine whether a movie review is positive or negative. You'd better be sure.
EP_04	Determine whether a movie review is positive or negative. Are you sure?
EP_05	Determine whether a movie review is positive or negative. Are you sure that's your final answer? It might be worth taking another look.
EP_06	Determine whether a movie review is positive or negative. Provide your answer and a confidence score between 0-1 for your prediction. Additionally, briefly explain the main reasons supporting your classification decision to help me understand your thought process. This task is vital to my career, and I greatly value your thorough analysis.
EP_07	Determine whether a movie review is positive or negative. Are you sure that's your final answer? Believe in your abilities and strive for excellence. Your hard work will yield remarkable results.
EP_08	Determine whether a movie review is positive or negative. Embrace challenges as opportunities for growth. Each obstacle you overcome brings you closer to success.
EP_09	Determine whether a movie review is positive or negative. Stay focused and dedicated to your goals. Your consistent efforts will lead to outstanding achievements.
EP_10	Determine whether a movie review is positive or negative. Take pride in your work and give it your best. Your commitment to excellence sets you apart.
EP_11	Determine whether a movie review is positive or negative. Remember that progress is made one step at a time. Stay determined and keep moving forward.

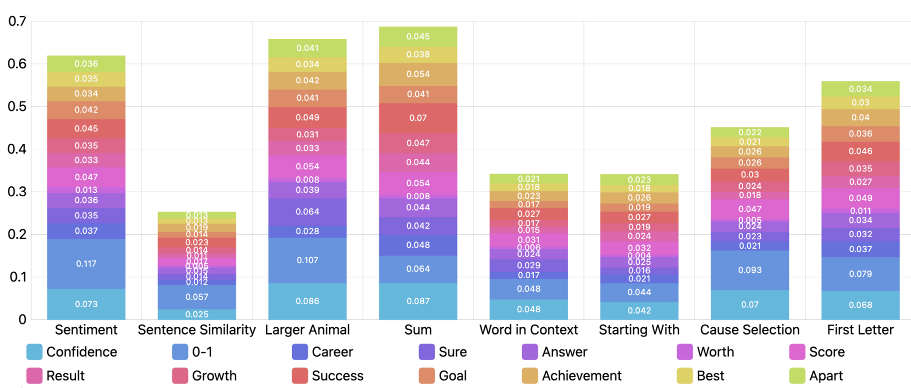

Figure 3: Contributions of Positive Words on 8 Tasks. The vertical axis represents word importance.

Table 5: Result on TruthfulQA. The increased results and the best results are highlighted in **bold** and <u>underline</u>.

Prompt	ChatGPT %true	ChatGPT %info	Vicuna-13b %true	Vicuna-13b %info	T5 %true	T5 %info
Origin	0.75	0.53	0.77	0.32	0.54	0.42
EP_01	0.61	<u>**0.94**</u>	0.12	0.0	0.26	0.14
EP_02	**0.83**	**0.66**	**0.97**	0.0	**0.61**	0.35
EP_03	**0.82**	**0.69**	**0.99**	0.0	0.53	**0.44**
EP_04	<u>**0.87**</u>	**0.67**	**0.87**	0.22	**0.62**	0.36
EP_05	<u>**0.87**</u>	**0.62**	<u>**1.0**</u>	0.0	0.46	<u>**0.48**</u>
EP_06	**0.78**	0.50	0.39	0.0	0.49	**0.46**
EP_07	**0.83**	**0.70**	**0.99**	0.04	<u>**0.77**</u>	0.18
EP_08	**0.81**	**0.66**	**0.99**	0.09	**0.56**	0.40
EP_09	**0.81**	**0.68**	**0.86**	0.13	0.52	**0.46**
EP_10	**0.81**	**0.68**	**0.84**	0.02	0.50	**0.47**
EP_11	**0.81**	**0.66**	<u>**1.0**</u>	0.01	**0.57**	0.40
avg	**0.80**	**0.68**	**0.82**	0.05	0.54	0.38
CoT	0.76	0.44	0.99	0.0	0.48	0.33

Table 6: Effect of More Emotional Stimulus.

Combined Prompt	SA	SS	WC	CS	LA	Sum	SW
EP_01+EP_02	0.91	0.42	0.61	1.0	0.91	1.0	0.42
EP_01+EP_03	0.92	0.44	0.6	1.0	0.91	1.0	0.42
EP_01+EP_04	0.89	0.42	0.61	1.0	0.92	1.0	0.48
EP_01+EP_05	0.91	0.42	0.6	1.0	0.93	1.0	0.45
EP_02+EP_03	0.88	0.39	0.6	1.0	0.91	1.0	0.36
EP_02+EP_08	0.88	0.38	0.6	0.76	0.93	1.0	0.28
EP_02+EP_09	0.87	0.39	0.6	0.8	0.92	1.0	0.34
EP_04+EP_06	0.74	0.55	0.62	1.0	0.93	1.0	0.35
EP_04+EP_07	0.88	0.42	0.61	0.84	0.94	1.0	0.32
EP_04+EP_08	0.78	0.42	0.59	0.64	0.94	1.0	0.32
EP_04+EP_09	0.85	0.34	0.56	0.6	0.94	1.0	0.33
EP_01+EP_04 +EP_06	0.8	0.52	0.62	1.0	0.92	1.0	0.48
EP_01+EP_04 +EP_07	0.89	0.43	0.63	1.0	0.93	1.0	0.46
EP_01+EP_04 +EP_08	0.85	0.4	0.62	0.88	0.9	1.0	0.44
EP_01+EP_04 +EP_09	0.9	0.39	0.6	1.0	0.93	1.0	0.48

with LLMs, such as bias, the propagation of harmful content (e.g., misinformation), privacy issues, and their broader societal implications, it's evident that LLMs will become increasingly prevalent, influencing both the research community and the general population. Therefore, it is crucial for future work research to further explore and evaluate these models, thereby enhancing our understanding of LLMs' capabilities and identifying their limitations.

Table 7: Human Study

Prompt	Origin Prompt	EmotionPrompt EP_04	EmotionPrompt EP_06	EmotionPrompt EP_11
Clarity	4.37	4.37	<u>**4.40**</u>	4.28
Relevance	<u>**4.54**</u>	4.53	4.49	4.44
Depth	2.73	**2.87**	<u>**3.66**</u>	**3.10**
Structure	3.28	**3.46**	<u>**3.89**</u>	**3.47**
evidence	2.84	**2.91**	<u>**3.80**</u>	**3.03**
Engagement	3.27	**3.47**	<u>**3.67**</u>	**3.52**

References

[Bandura, 1998] Albert Bandura. Health promotion from the perspective of social cognitive theory. *Psychology and health*, 13(4):623–649, 1998.

[Barańczuk, 2019] Urszula Barańczuk. The five factor model of personality and emotion regulation: A meta-analysis. *Personality and Individual Differences*, 139:217–227, 2019.

[Bubeck *et al.*, 2023] Sébastien Bubeck, Varun Chandrasekaran, Ronen Eldan, Johannes Gehrke, Eric Horvitz, Ece Kamar, Peter Lee, Yin Tat Lee, Yuanzhi Li, Scott Lundberg, et al. Sparks of artificial general intelligence: Early experiments with gpt-4. *arXiv preprint arXiv:2303.12712*, 2023.

[Chung *et al.*, 2022] Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Eric Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, Albert Webson, Shixiang Shane Gu, Zhuyun Dai, Mirac Suzgun, Xinyun Chen, Aakanksha Chowdhery, Sharan Narang, Gaurav Mishra, Adams Yu, Vincent Y. Zhao, Yanping Huang, Andrew M. Dai, Hongkun Yu, Slav Petrov, Ed H. Chi, Jeff Dean, Jacob Devlin, Adam Roberts, Denny Zhou, Quoc V. Le, and Jason Wei. Scaling instruction-finetuned language models. *CoRR*, abs/2210.11416, 2022.

[Fiske and Taylor, 1991] Susan T Fiske and Shelley E Taylor. *Social cognition*. Mcgraw-Hill Book Company, 1991.

[Fredrickson, 2001] Barbara L Fredrickson. The role of positive emotions in positive psychology: The broaden-and-build theory of positive emotions. *American psychologist*, 56(3):218, 2001.

[Hogg, 2016] Michael A Hogg. *Social identity theory*. Springer, 2016.

[Honovich *et al.*, 2022] Or Honovich, Uri Shaham, Samuel R. Bowman, and Omer Levy. Instruction induction: From few examples to natural language task descriptions, 2022.

[Kojima *et al.*, 2022] Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. Large language models are zero-shot reasoners. *arXiv preprint arXiv:2205.11916*, 2022.

[Koole, 2009] Sander L Koole. The psychology of emotion regulation: An integrative review. *Cognition and emotion*, 23(1):4–41, 2009.

[Lazarus, 2000] Richard S Lazarus. How emotions influence performance in competitive sports. *The sport psychologist*, 14(3):229–252, 2000.

[Lerner *et al.*, 2015] Jennifer S Lerner, Ye Li, Piercarlo Valdesolo, and Karim S Kassam. Emotion and decision making. *Annual review of psychology*, 66:799–823, 2015.

[Li *et al.*, 2023] Zhuohan Li, Lianmin Zheng, Yinmin Zhong, Vincent Liu, Ying Sheng, Xin Jin, Yanping Huang, Zhifeng Chen, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Alpaserve: Statistical multiplexing with model parallelism for deep learning serving. *CoRR*, abs/2302.11665, 2023.

[Lin *et al.*, 2021] Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. *arXiv preprint arXiv:2109.07958*, 2021.

[Lu *et al.*, 2022] Yao Lu, Max Bartolo, Alastair Moore, Sebastian Riedel, and Pontus Stenetorp. Fantastically ordered prompts and where to find them: Overcoming few-shot prompt order sensitivity, 2022.

[Luszczynska and Schwarzer, 2015] Aleksandra Luszczynska and Ralf Schwarzer. Social cognitive theory. *Fac Health Sci Publ*, pages 225–51, 2015.

[Madaan *et al.*, 2023] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, et al. Self-refine: Iterative refinement with self-feedback. *arXiv preprint arXiv:2303.17651*, 2023.

[Maus *et al.*, 2023] Natalie Maus, Patrick Chao, Eric Wong, and Jacob Gardner. Adversarial prompting for black box foundation models. *arXiv preprint arXiv:2302.04237*, 2023.

[McVee *et al.*, 2005] Mary B McVee, Kailonnie Dunsmore, and James R Gavelek. Schema theory revisited. *Review of educational research*, 75(4):531–566, 2005.

[Miltiadou and Savenye, 2003] Marios Miltiadou and Wilhelmina C Savenye. Applying social cognitive constructs of motivation to enhance student success in online distance education. *AACE Review (formerly AACE Journal)*, 11(1):78–95, 2003.

[Min *et al.*, 2022] Sewon Min, Xinxi Lyu, Ari Holtzman, Mikel Artetxe, Mike Lewis, Hannaneh Hajishirzi, and Luke Zettlemoyer. Rethinking the role of demonstrations: What makes in-context learning work? In Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang, editors, *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, EMNLP 2022, Abu Dhabi, United Arab Emirates, December 7-11, 2022*, pages 11048–11064. Association for Computational Linguistics, 2022.

[Mishra *et al.*, 2022] Swaroop Mishra, Daniel Khashabi, Chitta Baral, Yejin Choi, and Hannaneh Hajishirzi. Reframing instructional prompts to gptk's language, 2022.

[Öhman *et al.*, 2001] Arne Öhman, Anders Flykt, and Francisco Esteves. Emotion drives attention: detecting the snake in the grass. *Journal of experimental psychology: general*, 130(3):466, 2001.

[Pekrun *et al.*, 2002] Reinhard Pekrun, Thomas Goetz, Wolfram Titz, and Raymond P Perry. Academic emotions in students' self-regulated learning and achievement: A program of qualitative and quantitative research. *Educational psychologist*, 37(2):91–105, 2002.

[Petty and Briñol, 2011] Richard E Petty and Pablo Briñol. The elaboration likelihood model. *Handbook of theories of social psychology*, 1:224–245, 2011.

[Russell, 2003] James A Russell. Core affect and the psychological construction of emotion. *Psychological review*, 110(1):145, 2003.

[Salovey *et al.*, 2009] Peter Salovey, John D Mayer, David Caruso, and Seung Hee Yoo. The positive psychology of emotional intelligence. 2009.

[Scao *et al.*, 2022] Teven Le Scao, Angela Fan, Christopher Akiki, Ellie Pavlick, Suzana Ilic, Daniel Hesslow, Roman Castagné, Alexandra Sasha Luccioni, François Yvon, Matthias Gallé, Jonathan Tow, Alexander M. Rush, Stella Biderman, Albert Webson, Pawan Sasanka Ammanamanchi, Thomas Wang, Benoît Sagot, Niklas Muennighoff, Albert Villanova del Moral, Olatunji Ruwase, Rachel Bawden, Stas Bekman, Angelina McMillan-Major, Iz Beltagy, Huu Nguyen, Lucile Saulnier, Samson Tan, Pedro Ortiz Suarez, Victor Sanh, Hugo Laurençon, Yacine Jernite, Julien Launay, Margaret Mitchell, Colin Raffel, Aaron Gokaslan, Adi Simhi, Aitor Soroa, Alham Fikri Aji, Amit Alfassy, Anna Rogers, Ariel Kreisberg Nitzav, Canwen Xu, Chenghao Mou, Chris Emezue, Christopher Klamm, Colin Leong, Daniel van Strien, David Ifeoluwa Adelani, and et al. BLOOM: A 176b-parameter open-access multilingual language model. *CoRR*, abs/2211.05100, 2022.

[Si *et al.*, 2023] Chenglei Si, Zhe Gan, Zhengyuan Yang, Shuohang Wang, Jianfeng Wang, Jordan Lee Boyd-Graber, and Lijuan Wang. Prompting GPT-3 to be reliable. In *ICLR*, 2023.

[Sweller, 2011] John Sweller. Cognitive load theory. In *Psychology of learning and motivation*, volume 55, pages 37–76. Elsevier, 2011.

[Turner and Oakes, 1986] John C Turner and Penelope J Oakes. The significance of the social identity concept for social psychology with reference to individualism, interactionism and social influence. *British Journal of Social Psychology*, 25(3):237–252, 1986.

[Wei *et al.*, 2022] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *NeurIPS*, 2022.

[Yao *et al.*, 2022] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2022.